

Easy (10 questions)

1.

What does SDN stand for?

- A) Software Defined Networking
- B) Secure Data Network
- C) Simple Device Network
- D) Software Design Network

Answer: A) Software Defined Networking

Explanation: SDN stands for Software Defined Networking, a modern approach to network management.

2.

In SDN, which plane is separated from the data plane?

- A) Control plane
- B) User plane
- C) Management plane
- D) Security plane

Answer: A) Control plane

Explanation: SDN separates the control plane from the data (forwarding) plane.

3.

Which device controls the network in an SDN architecture?

- A) SDN Controller
- B) Router
- C) Switch
- D) Firewall

Answer: A) SDN Controller

Explanation: The SDN controller centrally manages the control plane functions.

4.

What protocol is commonly used as a southbound interface in SDN?

- A) HTTP
- B) OpenFlow
- C) FTP
- D) SMTP

Answer: B) OpenFlow

Explanation: OpenFlow is a widely used protocol for communication between controller and switches.

5.

Which SDN layer contains network applications like load balancing?

- A) Application layer
- B) Control layer
- C) Data layer
- D) Physical layer

Answer: A) Application layer

Explanation: Network applications reside in the application layer and communicate with the controller.

6.

Which of the following is NOT a benefit of SDN?

- A) Centralized control
- B) Manual device configuration

- C) Network programmability
- D) Dynamic network management

Answer: B) Manual device configuration

Explanation: SDN reduces manual configuration by automating control through software.

7.

Which layer in SDN architecture provides northbound APIs?

- A) Data plane
- B) Control plane
- C) Application plane
- D) Physical layer

Answer: B) Control plane

Explanation: The control plane exposes northbound APIs for applications to program the network.

8.

Which SDN component directly forwards packets?

- A) Controller
- B) Switch
- C) Application
- D) Database

Answer: B) Switch

Explanation: Switches execute forwarding decisions as instructed by the controller.

9.

What is the primary purpose of separating control and data planes?

- A) Simplify physical wiring
- B) Enable centralized control and flexibility
- C) Increase hardware costs
- D) Reduce software usage

Answer: B) Enable centralized control and flexibility

Explanation: Separation allows centralized management and easier network programmability.

10.

Which of the following is a common challenge of SDN?

- A) Controller failure impact
- B) Hardware incompatibility
- C) High manual configuration
- D) No network visibility

Answer: A) Controller failure impact

Explanation: If the controller fails, it may disrupt the entire network as it is centralized.

Medium (30 questions)

11.

How does SDN improve network scalability?

- A) By distributing control to multiple controllers

- B) By increasing manual configuration
- C) By removing all routers
- D) By reducing switch capacity

Answer: A) By distributing control to multiple controllers

Explanation: Distributed or hierarchical controllers improve scalability in large networks.

12.

Which SDN feature allows dynamic prioritization of traffic?

- A) QoS (Quality of Service)
- B) VLAN tagging
- C) Static routing
- D) Port mirroring

Answer: A) QoS (Quality of Service)

Explanation: SDN dynamically manages QoS policies to prioritize different traffic types.

13.

What does the SDN controller use to communicate with network devices?

- A) Northbound APIs
- B) Southbound APIs
- C) User interfaces
- D) Email protocols

Answer: B) Southbound APIs

Explanation: Southbound APIs like OpenFlow enable controllers to configure network devices.

14.

Which is a key reason SDN offers better reliability?

- A) Centralized monitoring and failover mechanisms
- B) Increased physical cabling
- C) Manual switch management
- D) Reducing network visibility

Answer: A) Centralized monitoring and failover mechanisms

Explanation: Centralized control enables fast detection and rerouting around failures.

15.

Which term describes the global network view in SDN?

- A) Network abstraction
- B) Network topology
- C) Network segmentation
- D) Network slicing

Answer: A) Network abstraction

Explanation: SDN abstracts the physical network into a logical global view.

16.

What is a common SDN southbound protocol besides OpenFlow?

- A) NETCONF
- B) FTP

- C) HTTP
- D) SMTP

Answer: A) NETCONF

Explanation: NETCONF is another protocol used to program network devices in SDN.

17.

Which SDN feature helps prevent unauthorized changes to network configurations?

- A) Access control and policy enforcement
- B) Static IP addressing
- C) VLANs
- D) Port forwarding

Answer: A) Access control and policy enforcement

Explanation: Centralized controllers enforce security policies and access control.

18.

In SDN, what does the northbound API provide?

- A) Interface for applications to communicate with the controller
- B) Interface for switches to forward data
- C) Interface for device firmware updates
- D) Interface for hardware installation

Answer: A) Interface for applications to communicate with the controller

Explanation: Northbound APIs allow apps to request network services and resources.

19.

Which SDN architecture component has the "global network view"?

- A) SDN Controller
- B) Router
- C) Switch
- D) Firewall

Answer: A) SDN Controller

Explanation: The controller maintains a centralized global view of the network.

20.

What is a major opportunity offered by SDN?

- A) Rapid innovation and service deployment
- B) Increased manual device configuration
- C) Reduced network visibility
- D) Static network behavior

Answer: A) Rapid innovation and service deployment

Explanation: SDN enables quick programming and deployment of new network services.

21.

What is a significant challenge when implementing SDN?

- A) Controller redundancy and fault tolerance
- B) Lack of protocols
- C) Manual routing
- D) Slow hardware

Answer: A) Controller redundancy and fault tolerance

Explanation: Controllers must be resilient to avoid single points of failure.

22.

Which layer translates application requests into flow rules?

- A) Control plane
- B) Data plane
- C) Application layer
- D) Physical layer

Answer: A) Control plane

Explanation: The control plane programs network devices with flow rules based on app requests.

23.

Which SDN capability aids service provider networks most?

- A) Automated provisioning and management
- B) Manual switch configuration
- C) Fixed routing paths
- D) Reduced bandwidth

Answer: A) Automated provisioning and management

Explanation: SDN automates service provider network management for faster and efficient operations.

24.

What does SDN use to monitor network device health?

- A) Controller's global view and telemetry data
- B) Manual ping commands
- C) Static IP lists
- D) Hardware switches only

Answer: A) Controller's global view and telemetry data

Explanation: Controllers collect telemetry to detect failures and optimize routing.

25.

Which term describes dividing a physical network into multiple logical networks?

- A) Network virtualization
- B) Static routing
- C) VLAN pruning
- D) Access control

Answer: A) Network virtualization

Explanation: SDN facilitates network virtualization for logical segmentation.

26.

Which is NOT a typical SDN challenge?

- A) Legacy device compatibility
- B) Centralized security vulnerabilities
- C) Inability to automate networks
- D) Standardization issues

Answer: C) Inability to automate networks

Explanation: SDN's strength is network automation, not inability.

27.

What allows SDN to adapt to changing network traffic patterns?

- A) Dynamic flow control via the controller
- B) Static ACLs
- C) Fixed routing tables
- D) Manual reconfiguration

Answer: A) Dynamic flow control via the controller

Explanation: Controllers dynamically adjust flow rules for optimal traffic management.

28.

Which of the following is a northbound API feature?

- A) RESTful API for network programmability
- B) Firmware updates to switches
- C) Forwarding packets
- D) Physical cable management

Answer: A) RESTful API for network programmability

Explanation: Northbound APIs are often RESTful interfaces used by applications.

29.

How does SDN contribute to network security?

- A) Centralized policy enforcement and monitoring
- B) Manual firewall configuration only
- C) Disabling encryption
- D) Random device access

Answer: A) Centralized policy enforcement and monitoring

Explanation: SDN enforces consistent security policies network-wide from a central point.

30.

Which SDN challenge involves interoperability issues?

- A) Standardization and protocol diversity
- B) Physical wiring complexity
- C) Lack of user interface
- D) Hardware overheating

Answer: A) Standardization and protocol diversity

Explanation: Multiple competing SDN protocols can cause interoperability problems.

31.

Which SDN characteristic aids cloud data centers?

- A) Network programmability and automation
- B) Fixed routing only
- C) Manual IP assignment
- D) Static network maps

Answer: A) Network programmability and automation

Explanation: Cloud data centers benefit from SDN's dynamic network control.

32.

What is the main function of an SDN controller?

- A) Manage forwarding behavior of switches
- B) Manufacture switches
- C) Monitor physical cables
- D) Encrypt data only

Answer: A) Manage forwarding behavior of switches

Explanation: Controllers send flow rules that control packet forwarding.

33.

What is the typical format of an OpenFlow flow entry?

- A) Match fields + actions + counters
- B) IP addresses only
- C) MAC addresses only
- D) Physical device IDs only

Answer: A) Match fields + actions + counters

Explanation: Flow entries include packet matching rules, actions, and statistics counters.

34.

Which SDN layer interfaces directly with hardware switches?

- A) Data plane

- B) Control plane
- C) Application plane
- D) Management plane

Answer: A) Data plane

Explanation: The data plane contains the physical switches forwarding packets.

35.

Which feature of SDN allows easy rollback of network configurations?

- A) Centralized version control
- B) Manual CLI commands
- C) Fixed routing tables
- D) Static VLANs

Answer: A) Centralized version control

Explanation: Controllers can manage configurations with versioning and rollback support.

36.

What does SDN enable for multi-tenant environments?

- A) Network slicing and isolation
- B) Physical separation of hardware
- C) Manual VLAN configuration only
- D) Static IP allocation only

Answer: A) Network slicing and isolation

Explanation: SDN supports creating isolated logical networks for tenants.

37.

Which protocol is often used for secure controller-to-device communication?

- A) TLS (Transport Layer Security)
- B) FTP
- C) HTTP
- D) SMTP

Answer: A) TLS (Transport Layer Security)

Explanation: TLS encrypts controller-device communication to secure SDN control channels.

38.

What is the purpose of a southbound API?

- A) Control communication from controller to network devices
- B) Interface for user apps
- C) Firmware updates only
- D) Cable diagnostics

Answer: A) Control communication from controller to network devices

Explanation: Southbound APIs manage device configuration and forwarding rules.

39.

Which of the following best describes SDN's data plane?

- A) Packet forwarding and processing
- B) Application logic

- C) Network monitoring only
- D) Software updates

Answer: A) Packet forwarding and processing

Explanation: The data plane is responsible for forwarding packets based on flow rules.

40.

How does SDN help service providers reduce operational costs?

- A) Automating provisioning and management
- B) Increasing manual device operations
- C) Using expensive proprietary hardware
- D) Disabling monitoring tools

Answer: A) Automating provisioning and management

Explanation: Automation reduces manual effort and speeds up service deployment.

Hard (10 questions)

41.

What SDN controller architecture can mitigate the single point of failure problem?

- A) Distributed or hierarchical controllers
- B) Single monolithic controller
- C) Manual routing protocols

D) Static routing tables

Answer: A) Distributed or hierarchical controllers

Explanation: Multiple controllers share the control load and provide redundancy.

42.

Which OpenFlow message is used by the controller to add a new flow entry?

- A) Flow Mod
- B) Packet In
- C) Echo Request
- D) Hello

Answer: A) Flow Mod

Explanation: The Flow Mod message programs flow entries in switches.

43.

Which challenge is most critical when deploying SDN at ISP scale?

- A) Multi-domain orchestration and interoperability
- B) Cable length issues
- C) Power consumption of switches
- D) User authentication complexity

Answer: A) Multi-domain orchestration and interoperability

Explanation: ISPs need SDN controllers to coordinate across multiple network domains.

44.

How does SDN ensure consistency in distributed controllers?

- A) Using consensus algorithms like RAFT or Paxos
- B) Manual sync between controllers
- C) Disabling controller redundancy
- D) Static routing updates

Answer: A) Using consensus algorithms like RAFT or Paxos

Explanation: Distributed controllers use consensus protocols to maintain consistent state.

45.

Which SDN security risk arises from centralized control?

- A) Controller compromise affecting entire network
- B) Switch hardware failure only
- C) Cable damage
- D) IP address exhaustion

Answer: A) Controller compromise affecting entire network

Explanation: Attacking the controller can disrupt or manipulate network-wide traffic.

46.

In SDN, what does the controller do upon receiving a Packet-In message?

- A) Decides how to handle unknown packets and installs flow entries
- B) Forwards packet automatically
- C) Drops the packet
- D) Encrypts the packet

Answer: A) Decides how to handle unknown packets and installs flow entries

Explanation: Packet-In messages notify the controller about packets without matching flows.

47.

Which QoS technique is often easier to implement in SDN compared to traditional networks?

- A) Dynamic flow-based QoS adjustments
- B) Static priority queues only
- C) Manual VLAN tagging
- D) Fixed bandwidth allocation

Answer: A) Dynamic flow-based QoS adjustments

Explanation: SDN dynamically adjusts QoS policies based on real-time traffic analysis.

48.

Which element is crucial for SDN controller scalability?

- A) Horizontal scaling with multiple controllers
- B) Single core processing
- C) Manual failover processes
- D) Fixed hardware limits

Answer: A) Horizontal scaling with multiple controllers

Explanation: Multiple controllers working together ensure scalability and reliability.

49.

What is the primary role of an SDN northbound API?

- A) Provide abstraction for application programming of the network
- B) Update switch firmware
- C) Manage physical cables
- D) Encrypt traffic flows

Answer: A) Provide abstraction for application programming of the network

Explanation: Northbound APIs enable applications to interact with and program the network.

50.

How does SDN assist in compliance and auditing?

- A) Centralized logging and policy enforcement
- B) Manual audit trails only
- C) Disabling logs
- D) No impact on compliance

Answer: A) Centralized logging and policy enforcement

Explanation: SDN centralizes network policies and logs, simplifying auditing and compliance.